Year 10 Higher

REVISION PACK

Algebra 1

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

Q	Working	Answer
substitution into v = u + at		
1.1	$v = 3 + (-4) \times 2$ $= 3 - 8$ $= -5$	$v = -5$
1.2	$v = 10 + (-3) \times 6$ $= 10 - 18$ $= -8$	$v = -8$
1.3	$v = -4 + 5 \times 3$ $= -4 + 15$ $= 11$	$v = 11$
1.4	$v = 20 + (-6) \times 3$ $= 20 - 18$ $= 2$	$v = 2$
1.5	$v = -2 + (-3) \times 4$ $= -2 - 12$ $= -14$	$v = -14$
1.6	$v = 15 + (-2.5) \times 4$ $= 15 - 10$ $= 5$	$v = 5$
substitution with a square		
1.7	$E = \frac{1}{2} \times 6 \times (-5)^2$ $= \frac{1}{2} \times 6 \times 25$ $= 75$	$E = 75$
1.8	$E = \frac{1}{2} \times 10 \times (-2)^2$ $= \frac{1}{2} \times 10 \times 4$ $= 20$	$E = 20$

1.9	$E = \frac{1}{2} \times 8 \times (-3)^2$ $= \frac{1}{2} \times 8 \times 9$ $= 36$	$E = 36$
1.10	$E = \frac{1}{2} \times 4 \times (-6)^2$ $= \frac{1}{2} \times 4 \times 36$ $= 72$	$E = 72$
1.11	$E = \frac{1}{2} \times 12 \times (-1)^2$ $= \frac{1}{2} \times 12 \times 1$ $= 6$	$E = 6$
1.12	$E = \frac{1}{2} \times 3 \times (-4)^2$ $= \frac{1}{2} \times 3 \times 16$ $= 24$	$E = 24$
simplifying expressions		
1.13		**(a)** $4x + 3y$ **(b)** $8a^4b^3$
1.14		**(a)** $6m + 4n$ **(b)** $18x^3y^3$
1.15		**(a)** $5c - 4d$ **(b)** $10p^3q^6$
1.16		**(a)** $10x + 4y$ **(b)** $18a^3b^4$
1.17		**(a)** $-3m + 7n$ **(b)** $14x^5y^3$
1.18		**(a)** $7p + 3q$ **(b)** $20a^4b^3$

Algebra 1 · Solving equations (Lesson 2.4)

Q	Working	Answer
unknowns on both sides		

2.1	$4x - 5 = 11$ $4x = 16$ $x = 4$	$x = 4$
2.2	$4x + 4 = 24$ $4x = 20$ $x = 5$	$x = 5$
2.3	$3x - 3 = 12$ $3x = 15$ $x = 5$	$x = 5$
2.4	$4x + 2 = -10$ $4x = -12$ $x = -3$	$x = -3$
2.5	$6x - 9 = 15$ $6x = 24$ $x = 4$	$x = 4$
2.6	$3x + 8 = 23$ $3x = 15$ $x = 5$	$x = 5$
equations with fractions		
2.7	$2x + 5 = 21$ $2x = 16$ $x = 8$	$x = 8$
2.8	$4x - 3 = 25$ $4x = 28$ $x = 7$	$x = 7$
2.9	$5x + 2 = 12$ $5x = 10$ $x = 2$	$x = 2$
2.10	$3x + 4 = 10$ $3x = 6$ $x = 2$	$x = 2$

2.11	$2x - 1 = 15$ $2x = 16$ $x = 8$	$x = 8$
2.12	$6x + 3 = 15$ $6x = 12$ $x = 2$	$x = 2$
equations with brackets		
2.13	$8x - 4 = 3x + 21$ $5x = 25$ $x = 5$	$x = 5$
2.14	$5x - 10 = 4x + 6$ $1x = 16$ $x = 16$	$x = 16$
2.15	$9x + 6 = 4x + 36$ $5x = 30$ $x = 6$	$x = 6$
2.16	$8x - 6 = 3x + 9$ $5x = 15$ $x = 3$	$x = 3$
2.17	$6x - 6 = 4x + 8$ $2x = 14$ $x = 7$	$x = 7$
2.18	$8x + 4 = 6x + 2$ $2x = -2$ $x = -1$	$x = -1$